Math 105L In Class 1 Practice Solutions

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Functions

1. Find the domain and range of the following functions:

   (a) $f(x) = \frac{1}{\sqrt{16-x^2}}$

   Domain: $-4 < x < 4$, Range: $y \geq \frac{1}{4}$

   (b) $g(x) = \frac{3x-1}{x+2}$

   Domain: $x \neq -2$, Range: $y \neq 3$

   (c) $h(x) = \frac{4}{x^2+2}$

   Domain: All real numbers, Range: $0 < y \leq 2$

   (d) $k(x) = \sqrt{\frac{x-1}{x+3}}$ (Hint: You'll need to know where the fraction is positive/negative.)

   Domain: $x < -3$ or $x \geq 1$, Range: $0 \leq y < 1$ or $y > 1$

   (e) $m(x) = 5 - |2x - 3|$

   Domain: All real numbers, Range: $y \leq 5$

   (f) $p(x) = \sqrt{x^2 - 9}$

   Domain: $x \leq -3$ or $x \geq 3$, Range: $y \geq 0$

   (g) $r(x) = \sqrt{9 - |x|}$

   Domain: $-9 \leq x \leq 9$, Range: $0 \leq y \leq 3$

2. Let $V = f(t)$ be the volume of gasoline (in gallons) remaining in a 15-gallon gas tank t hours after starting a road trip. The tank starts full, and the car runs out of gas after 5 hours.

   (a) What are the units of the domain and the range of f?

   Domain units: Hours. Range units: Gallons.

   (b) What is a suitable domain and range for f?

   Domain: $0 \leq t \leq 5$ hours. Range: $0 \leq V \leq 15$ gallons.

   (c) Describe in words what $f(2) - 3$ represents.

   3 gallons less than the volume of gasoline remaining in the tank 2 hours into the trip.

3. Let $T = g(m)$ be the temperature of an oven (in °F) m minutes after it is turned on to preheat.

   (a) Describe in words what $g(10)$ means, including units.

   The temperature of the oven 10 minutes after being turned on, in °F.

   (b) Describe in words the difference between $g(m + 15)$ and $g(m) + 15$.

   $g(m + 15)$ is the oven temperature 15 minutes later than time m (in °F). $g(m) + 15$ is 15 degrees Fahrenheit higher than the oven temperature at time m.

   (c) Write an expression for a temperature that is 20 degrees Fahrenheit lower than twice the oven temperature 5 minutes ago.

   $2g(m - 5) - 20$

4. A concert venue with a maximum capacity of 500 seats models its ticket sales using the function $N = C(p)$, where N is the number of tickets sold when the price per ticket is p dollars. Tickets are always priced between \$20 and \$100.

   (a) What would be a suitable domain for $C(p)$, including units?

   $20 \leq p \leq 100$ dollars.

   (b) What would be a suitable range for $C(p)$, including units and numerical type?

   Integers N such that $0 \leq N \leq 500$ tickets.

   (c) Describe in words what $C(50)$ represents.

   The number of tickets sold when the price per ticket is \$50.

5. Let $A = h(t)$ be the altitude (in meters) of a weather balloon t minutes after its launch.

   (a) Describe in words what $h(12)$ represents, including units.

   The altitude of the weather balloon 12 minutes after launch, in meters.

   (b) Describe in words what $h(t - 10)$ represents.

   The altitude of the weather balloon 10 minutes prior to time t, in meters.

   (c) Write an expression for an altitude that is 100 meters higher than half the balloon's altitude 15 minutes from now.

   $\frac{1}{2}h(t + 15) + 100$

6. A coffee shop calculates its daily revenue R (in dollars) as a function of n, the number of cups of coffee sold on that day, given by $R = R(n)$. The shop can make at most 600 cups of coffee per day, and each cup costs \$4.

   (a) What is a sensible domain for $R(n)$? What are its units?

   Integers n such that $0 \leq n \leq 600$ cups of coffee.

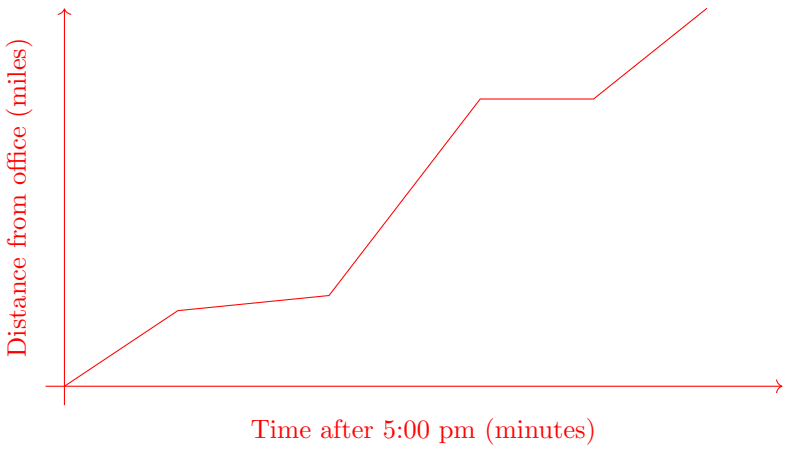
   (b) What is a sensible range for $R(n)$? What are its units?

   Multiples of 4 dollars, R, such that $0 \leq R \leq 2400$ dollars.

   (c) Write an expression for the revenue earned from selling 30 more cups of coffee than on a day when n cups were sold.

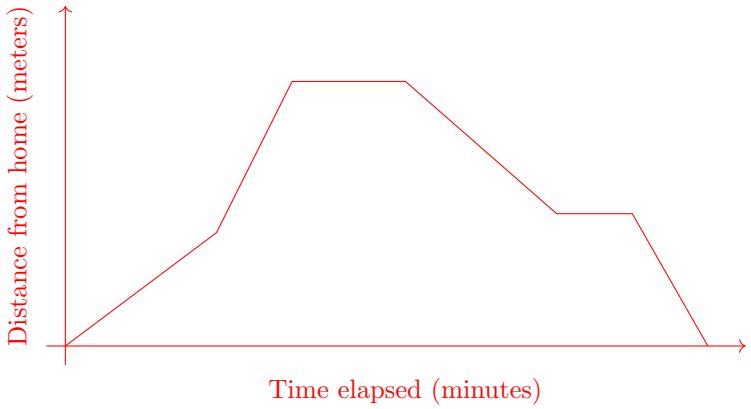
   $R(n + 30)$

7. You leave your office at 5:00 pm to drive home. For the first few minutes, you drive at a steady pace on local roads. You then get stuck in heavy stop-and-go traffic where you barely cover any distance. Once clear of traffic, you drive on the highway at a high, constant speed. Suddenly, you pull over at a gas station and stay there for a few minutes to fill up. Finally, you resume driving on the highway at a high speed until you reach home.
   Draw a graph of your distance from your office as a function of time. Take care to show distinct differences in slope between local driving, stop-and-go traffic, highway driving, and stopping for gas. Label your axes with variables and appropriate units.



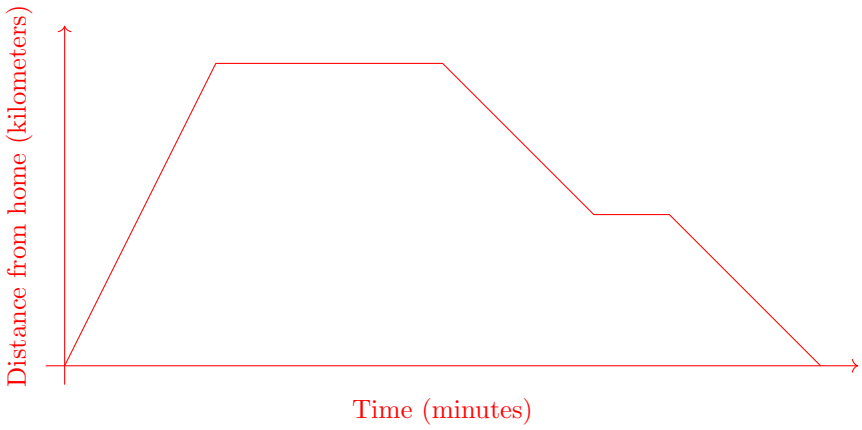
Ensure the graph has flat zero-slope sections for the gas station stop, a nearly flat section for heavy traffic, and a distinctly steeper slope for highway driving compared to local driving.

8. You take your dog for a walk starting from your house. You walk at a relaxed, steady pace away from home. Your dog suddenly spots a squirrel and sprints straight ahead away from home for a short distance. You stop to catch your breath for two minutes. Realizing you went too far, you walk back halfway home at a steady pace, stop briefly to talk to a neighbor, and then jog the remaining distance back home.
Draw a possible graph of your distance from home as a function of time. Label your axes with appropriate variables and units. Ensure differences between walking, sprinting, stopping, and jogging are clear.



Check that sprinting away is steeper than walking away, stopping creates horizontal line segments, walking halfway back has a negative slope, and jogging the rest of the way home is steeper than walking back.

9. You leave your home on a bicycle to go to the public library. You ride at a fast, constant speed until you arrive at the library. You spend 30 minutes inside studying. Upon leaving, you start heading back home at a slow, leisurely ride. Halfway home, you get a flat tire, so you stop for a few minutes to inspect it, and then walk the rest of the way home pushing your bike.
Draw a possible graph of your distance from home as a function of time. Take care to show the differences in speed between riding fast, riding slow, stopping, and walking. Label your axes with variables and units.



Verify that the initial ride to the library is steep, the library stay and tire inspection are horizontal flat lines, and walking home is less steep (slower) than riding slowly.

10.

(a) For the relationship $p - q^3 = 1$, is p a function of q? If so, what is the range? If not, why not?

Yes, p is a function of q. Range: All real numbers.

(b) For the relationship $p + |q| = 5$, is p a function of q? If so, what is the range? If not, why not?

Yes, p is a function of q. Range: $p \leq 5$.

(c) For the relationship $p^2 - q = 4$, is p a function of q? If so, what is the range? If not, why not?

No, p is not a function of q. For example, if $q = 0$, $p^2 = 4$, which gives $p = 2$ or $p = -2$ (two outputs for a single input).

(d) For the relationship $|p| + q = 3$, is p a function of q? If so, what is the range? If not, why not?

No, p is not a function of q. For example, if $q = 0$, $|p| = 3$, which gives $p = 3$ or $p = -3$ (two outputs for a single input).

Linear Functions

1. (a) Suppose $h(x)$ is a linear function. Find its equation given that its slope is -4 and that $h(3) = 2$.
$y - 2 = -4(x - 3)$, or $y = -4x + 14$.

(b) Suppose $k(x)$ is a linear function. Find its equation given that $k(-1) = 7$ and $k(3) = -1$.
Slope is $\frac{-1-7}{3-(-1)} = \frac{-8}{4} = -2$. So $y - 7 = -2(x + 1)$, or $y = -2x + 5$.

(c) Suppose that the line L is parallel to $y = -\frac{3}{4}x + 5$ and passes through the point $(-2, 4)$. Find its equation.
$m = -\frac{3}{4}$, so $y - 4 = -\frac{3}{4}(x + 2)$, or $y = -\frac{3}{4}x + \frac{5}{2}$.

(d) Find an equation for the line L that is perpendicular to $y = 2x - 3$ and passes through the point $(6, -1)$.
$m = -\frac{1}{2}$, so $y - (-1) = -\frac{1}{2}(x - 6)$, or $y = -\frac{1}{2}x + 2$.

(e) Find an equation for the line L that passes through the point $(4, 1)$ and is perpendicular to the line connecting $(1, 2)$ and $(3, 8)$.
Slope of the given line is $\frac{8-2}{3-1} = 3$. The perpendicular slope is $m = -\frac{1}{3}$. So $y - 1 = -\frac{1}{3}(x - 4)$, or $y = -\frac{1}{3}x + \frac{7}{3}$.

2. A school district models the average annual paper usage P (in reams of paper) per classroom as a linear function of the number of digital assignment days d per month, given by

$$P = -4d + 80.$$

If a district with 150 classrooms increases its digital assignment days by 3 days per month, how many total reams of paper does the model predict will be saved across the entire district over a 2-year period?
$\Delta d = 3$ days/month. So $\Delta P = -4 \times 3 = -12$ reams/classroom/year.
Over 2 years for 1 classroom: $12 \times 2 = 24$ reams.
For 150 classrooms over 2 years: $24 \times 150 = 3,600$ reams.
So, 3,600 reams of paper will be saved.

3. **The Strawberry Patch** In a commercial farm study, the following facts are known about growing strawberries:

   - If 15 thousand plants are planted per acre, each plant yields 2.4 pounds of strawberries per season.
   - For each additional thousand plants planted per acre, the yield decreases by 0.15 pounds per plant.

   Let x be the number of thousands of plants per acre, and let Y be the yield in pounds per plant.

   (a) What is Y if $x = 15$? Include units.

   2.4 pounds per plant.

   (b) What information in the problem tells you that Y is a linear function of x? What is its slope?

   The yield Y decreases by a constant amount (0.15 pounds per plant) for every unit increase in x (every additional 1,000 plants per acre). The slope is -0.15.

   (c) Find a formula for Y in terms of x.

   $Y - 2.4 = -0.15(x - 15)$, or $Y = -0.15x + 4.65$ pounds per plant.

4. A municipal water authority models daily residential water usage W (in gallons per person) based on the municipal water rate r (in dollars per 1,000 gallons) using the linear function

$$W = -14.5r + 110.$$

   The county has a population of 45,000 residents. If the water authority raises the rate r by \$0.80 per 1,000 gallons, how many total gallons of water are expected to be conserved across the entire county during a 30-day month?

   $\Delta r = 0.80$. So $\Delta W = -14.5 \times 0.80 = -11.6$ gallons/person/day.

   Over 30 days for 1 person: $11.6 \times 30 = 348$ gallons.

   For 45,000 residents over 30 days: $348 \times 45,000 = 15,660,000$ gallons.

   So, 15,660,000 gallons of water are expected to be conserved.

5. **Commuter Rail Ridership** A regional transit agency tracks ridership on a popular commuter line:

   - When the ticket price is \$12, the train averages 1,800 passengers per trip.
   - For every \$0.50 increase in ticket price, ridership drops by 40 passengers per trip.

   Let p be the ticket price in dollars, and let R be the number of passengers per trip.

   (a) What is R when $p = 12$? Include units.

   1,800 passengers.

   (b) Show that R is a linear function of p by finding its rate of change (slope). Include units for the slope.

   The rate of change is constant: Slope $= \frac{\Delta R}{\Delta p} = \frac{-40}{0.50} = -80$ passengers per dollar.

   (c) Find a formula for R in terms of p.

   $R - 1800 = -80(p - 12)$, or $R = -80p + 2760$ passengers.

6. A coffee roastery analyzes its carbon footprint and determines that roasting coffee at a temperature of 400°F results in 3.2 pounds of carbon emissions per bag of roasted coffee. For every 5°F reduction in roasting temperature, carbon emissions decrease by 0.15 pounds per bag.

   (a) Explain why the carbon emissions per bag, C (in pounds), is a linear function of roasting temperature T (in °F), and state its slope.

   Emissions decrease at a constant rate relative to temperature change. Slope $= \frac{\Delta C}{\Delta T} = \frac{-0.15}{-5} =$ 0.03 pounds per bag per °F.

   (b) Find a formula for C in terms of T.

   $C - 3.2 = 0.03(T - 400)$, or $C = 0.03T - 8.8$ pounds per bag.

   (c) If the roastery reduces its roasting temperature from 400°F to 380°F, how many total pounds of carbon emissions will be prevented if they produce 25,000 bags of coffee this year?

   $\Delta T = 380 - 400 = -20$°F.
   $\Delta C = 0.03 \times (-20) = -0.6$ pounds per bag.
   Total carbon prevented: $0.6 \times 25,000 = 15,000$ pounds.

7. (a) The electrostatic force F between two point charges is directly proportional to the product of their charges, q_1 and q_2, and inversely proportional to the square of the distance r between them. Write a formula for F in terms of q_1, q_2, and r.
   $F = \frac{kq_1q_2}{r^2}$

   (b) The aerodynamic drag force D acting on a vehicle is directly proportional to the frontal surface area A of the vehicle and directly proportional to the square of its speed v. Write a formula for D in terms of A and v.
   $D = kAv^2$

(c) The electrical resistance R of a uniform wire is directly proportional to its length L and inversely proportional to its cross-sectional area A. Write a formula for R in terms of L and A.

$R = \frac{kL}{A}$

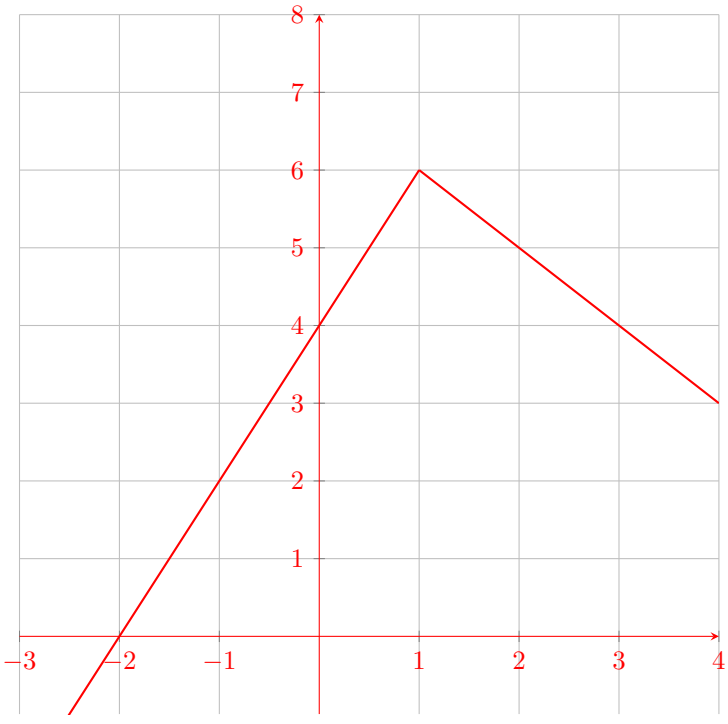
(d) The illuminance E on a surface produced by a light source is directly proportional to the luminous intensity I of the source and inversely proportional to the square of the distance d from the surface to the source. Write a formula for E in terms of I and d.

$E = \frac{kI}{d^2}$

8. Consider the following *piecewise linear* function:

$$f(t) = \begin{cases} 2t + 4, & t \leq 1, \\ -t + 7, & t > 1. \end{cases}$$

(a) Draw its graph.



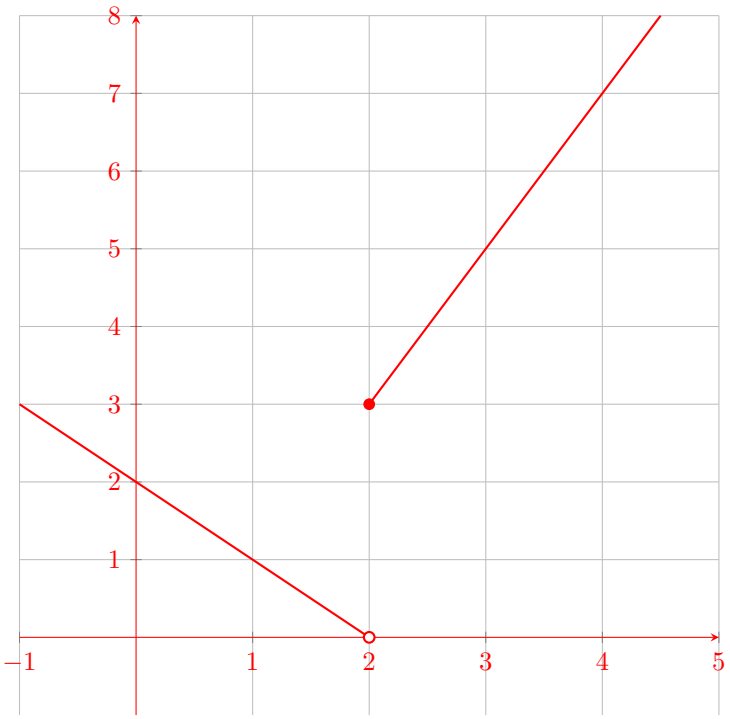
(b) Is $f(t)$ continuous at $t = 1$? Explain.

Yes, $f(t)$ is continuous at $t = 1$ because as t approaches 1 from both the left $(2(1) + 4 = 6)$ and the right $(-1 + 7 = 6)$, the function values approach 6, which equals $f(1)$.

9. Consider the following *piecewise linear* function:

$$g(t) = \begin{cases} -t+2, & t < 2, \\ 2t-1, & t \geq 2. \end{cases}$$

(a) Draw its graph.

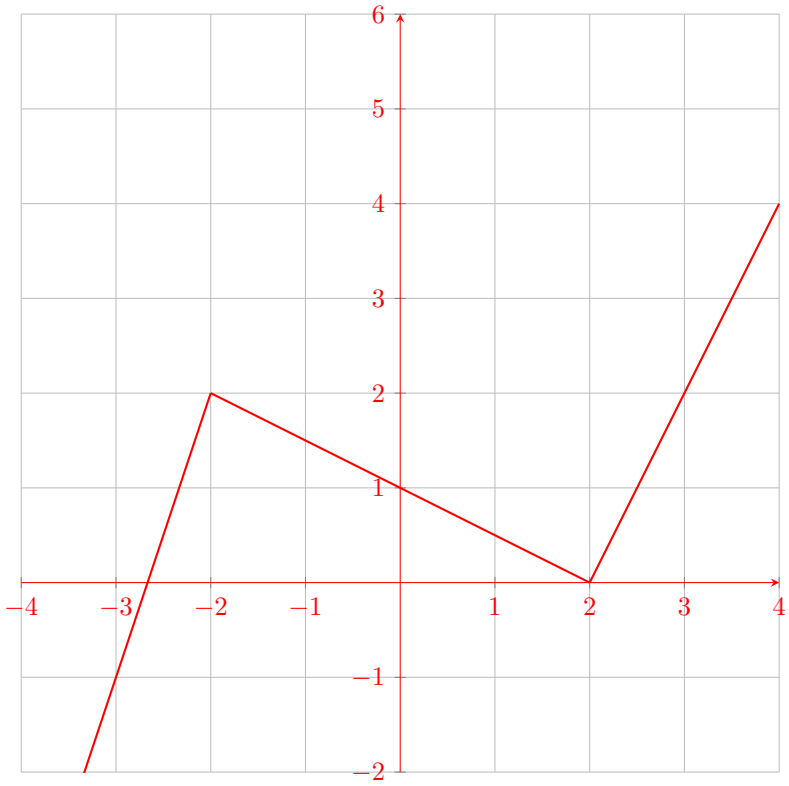


(b) Fill in the blank: At $t = 2$, the function $g(t)$ is not continuous.

10. Consider the following *piecewise linear* function:

$$h(t) = \begin{cases} 3t + 8, & t \leq -2, \\ -0.5t + 1, & -2 < t < 2, \\ 2t - 4, & t \geq 2. \end{cases}$$

(a) Draw its graph.



(b) Is $h(t)$ continuous everywhere? Explain what happens at $t = -2$ and $t = 2$.

Yes, $h(t)$ is continuous everywhere. At $t = -2$, both pieces evaluate to 2. At $t = 2$, both pieces evaluate to 0. Since the branches connect at these transition points, there are no jumps or gaps in the graph.

Power functions and Polynomials

1. (a) **True or False:** A polynomial of degree 6 must have exactly 5 turning points. Explain your reasoning.

   False. A polynomial of degree n has *at most* $n-1$ turning points. Therefore, a degree 6 polynomial can have up to 5 turning points, but it could also have fewer (such as 3 or 1 turning points).

   (b) **True or False:** For the polynomial $p(x) = -3x^5 + 7x^2 - 1$, both tails of the graph point downwards toward $-\infty$ as $x \to \pm\infty$. Explain your reasoning.

   False. The degree of $p(x)$ is odd ($n = 5$), which means its tails must point in *opposite* directions. Because the top coefficient is negative (-3), the left tail goes to ∞ as $x \to -\infty$, and the right tail goes to $-\infty$ as $x \to \infty$.

   (c) **True or False:** If a polynomial has an even degree and a positive top coefficient, then $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = \infty$. Explain your reasoning.

   True. For an even-degree polynomial, both tails go in the *same* direction. When the top coefficient is positive, both tails go to ∞.

   (d) **True or False:** Every cubic polynomial ($n = 3$) has at least one turning point. Explain your reasoning.

   False. A degree n polynomial can have at most $n-1$ turning points, but it does not have to reach that maximum. For example, the cubic polynomial $f(x) = x^3$ has 0 turning points.

   (e) **True or False:** The graph of $q(x) = -2x^4 + 5x^3 - x + 4$ has tails that point in opposite directions because the top coefficient is negative. Explain your reasoning.

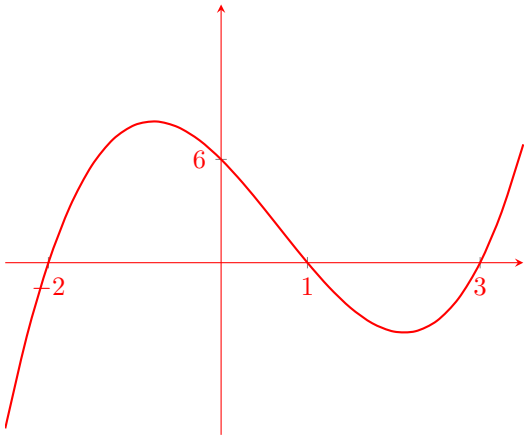
   False. Whether the tails go in the same or opposite directions depends on the *degree*, not the coefficient sign. Since the degree is even ($n = 4$), the tails go in the *same* direction. Because the top coefficient is negative (-2), both tails point downwards to $-\infty$.

   (f) **True or False:** If a polynomial graph exhibits 4 turning points, the degree of the polynomial must be at least 5. Explain your reasoning.

   True. Since a polynomial of degree n can have at most $n-1$ turning points, a polynomial with k turning points must satisfy $n - 1 \geq k$, or $n \geq k + 1$. For $k = 4$ turning points, the degree n must be at least $4 + 1 = 5$.

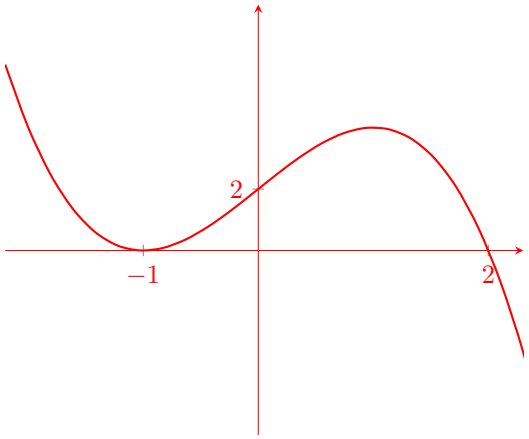
2. Without using a calculator or online tools, sketch the graphs of the following polynomials. Be sure to clearly label all x- and y-intercepts.

   (a) $y = (x + 2)(x - 1)(x - 3)$



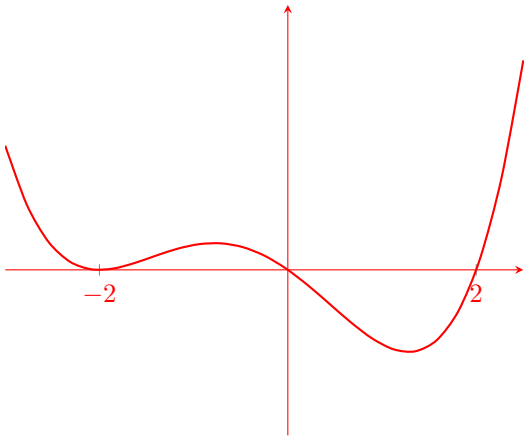
   x-intercepts: $x = -2, 1, 3$; y-intercept: $y = 6$.

(b) $y = -(x+1)^2(x-2)$



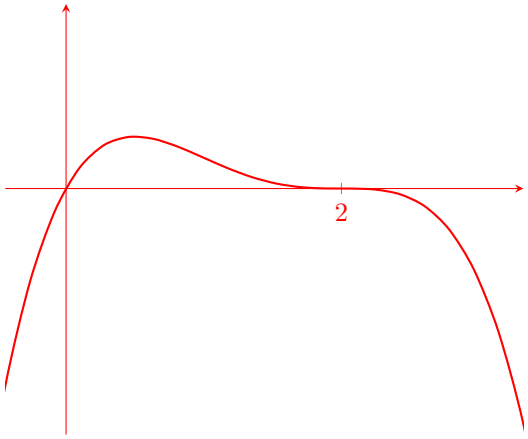
x-intercepts: $x = -1$ (touches/tangent), $x = 2$; y-intercept: $y = 2$.

(c) $y = \frac{1}{4}x(x+2)^2(x-2)$



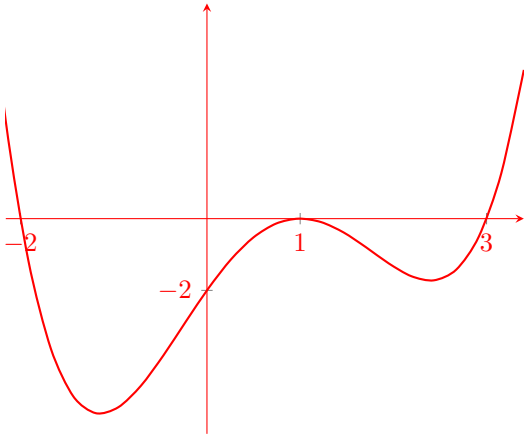
x-intercepts: $x = -2$ (touches/tangent), $x = 0, 2$; y-intercept: $y = 0$.

(d) $y = -x(x-2)^3$



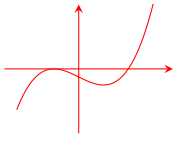
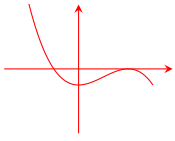
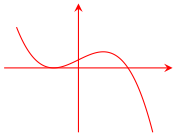
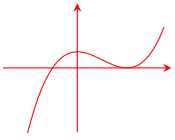
x-intercepts: $x = 0$, $x = 2$ (inflection/flattening); y-intercept: $y = 0$.

(e) $y = \frac{1}{3}(x+2)(x-1)^2(x-3)$

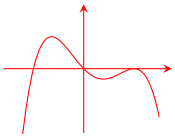
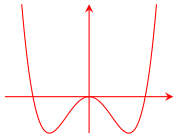
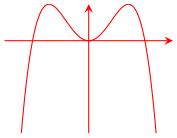
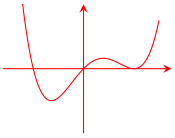


x-intercepts: $x = -2, 3$, $x = 1$ (touches/tangent); y-intercept: $y = -2$.

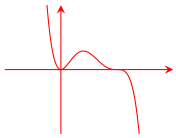
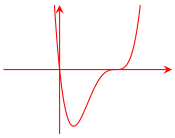
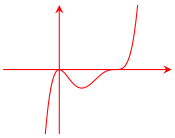
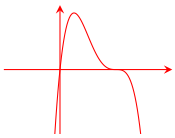
3. Without using a calculator or online tools, match each of the following polynomials with the correct graph.

$(x-2)^2(x+1)$	$-(x-2)^2(x+1)$	$(x-2)(x+1)^2$	$-(x-2)(x+1)^2$
(A)	**(B)**	**(C)**	**(D)**
			
C	B	D	A

4. Without using a calculator or online tools, match each of the following polynomials with the correct graph.

$x^2(x-2)(x+2)$	$-x^2(x-2)(x+2)$	$x(x-2)^2(x+2)$	$-x(x-2)^2(x+2)$
(A)	**(B)**	**(C)**	**(D)**
			
D	A	B	C

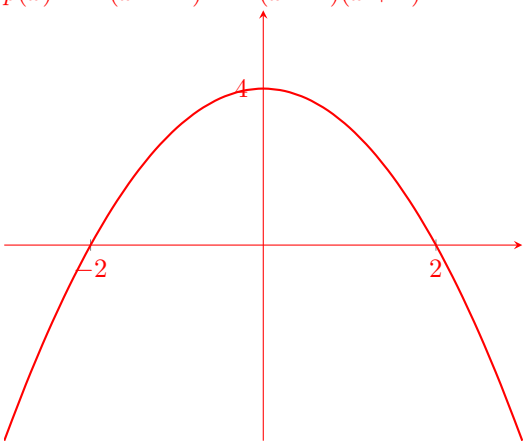
5. Without using a calculator or online tools, match each of the following polynomials with the correct graph.

$x(x-1)^3$	$-x(x-1)^3$	$x^2(x-1)^3$	$-x^2(x-1)^3$
(A)	**(B)**	**(C)**	**(D)**
			
D	A	C	B

6. Without using a calculator or online tools: by first factoring, sketch the graphs of the following polynomials. Be sure to label all of the intercepts.

(a) $p(x) = 4 - x^2$

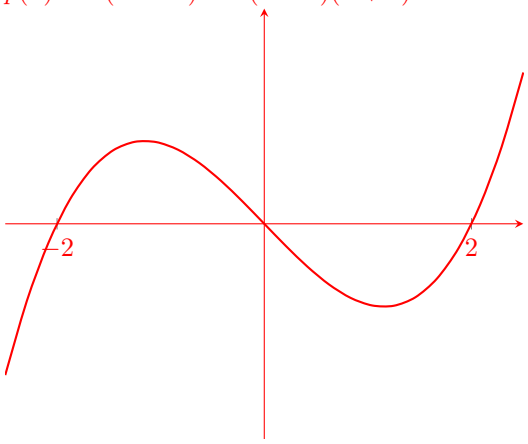
$p(x) = -(x^2 - 4) = -(x-2)(x+2)$



x-intercepts: $x = -2, 2$; y-intercept: $y = 4$.

(b) $p(x) = x^3 - 4x$

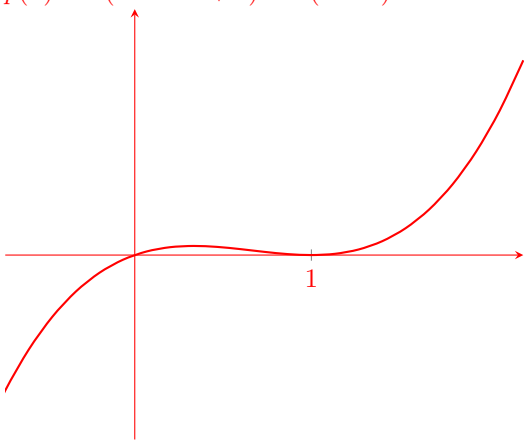
$p(x) = x(x^2 - 4) = x(x-2)(x+2)$



x-intercepts: $x = -2, 0, 2$; y-intercept: $y = 0$.

(c) $p(x) = x^3 - 2x^2 + x$

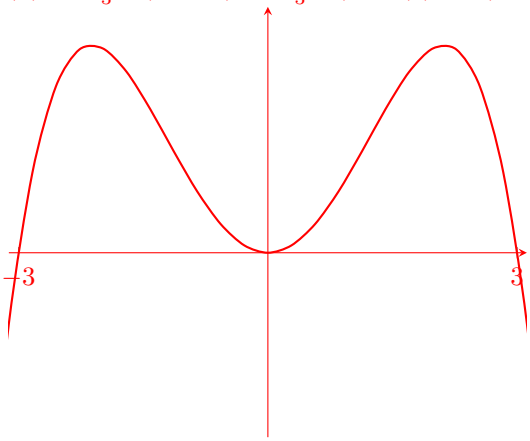
$p(x) = x(x^2 - 2x + 1) = x(x-1)^2$



x-intercepts: $x = 0$, $x = 1$ (touches/tangent); y-intercept: $y = 0$.

(d) $p(x) = -\frac{1}{3}(x^4 - 9x^2)$

$p(x) = -\frac{1}{3}x^2(x^2 - 9) = -\frac{1}{3}x^2(x-3)(x+3)$



x-intercepts: $x = -3, 3$, $x = 0$ (touches/tangent); y-intercept: $y = 0$.

Rational Functions

1. Find all of the intercepts and asymptotes of the following functions.

   (a) $r(x) = \frac{2x}{x^2-9}$
   x-intercept: $x = 0$
   y-intercept: $y = 0$
   Vertical Asymptotes: $x = -3, 3$
   Horizontal Asymptote Work: As $x \to \pm\infty$, the highest-degree terms dominate, so $r(x) \approx \frac{2x}{x^2} = \frac{2}{x} \to 0$. Thus, the Horizontal Asymptote is $y = 0$.

   (b) $r(x) = \frac{3x-6}{x+2}$
   x-intercept: $x = 2$
   y-intercept: $y = -3$
   Vertical Asymptote: $x = -2$
   Horizontal Asymptote Work: As $x \to \pm\infty$, the highest-degree terms dominate, so $r(x) \approx \frac{3x}{x} = 3$. Thus, the Horizontal Asymptote is $y = 3$.

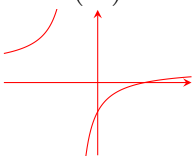
   (c) $r(x) = \frac{(x-2)(x+3)}{x-1}$
   x-intercepts: $x = -3, 2$
   y-intercept: $y = 6$
   Vertical Asymptote: $x = 1$
   Horizontal Asymptote Work: Expanding the numerator gives $r(x) = \frac{x^2+x-6}{x-1}$. As $x \to \pm\infty$, the highest-degree terms dominate, so $r(x) \approx \frac{x^2}{x} = x \to \pm\infty$. Thus, there is no Horizontal Asymptote.

   (d) $r(x) = \frac{4-x^2}{x^2+1}$
   x-intercepts: $x = -2, 2$
   y-intercept: $y = 4$
   Vertical Asymptotes: None
   Horizontal Asymptote Work: As $x \to \pm\infty$, the highest-degree terms dominate, so $r(x) \approx \frac{-x^2}{x^2} = -1$. Thus, the Horizontal Asymptote is $y = -1$.

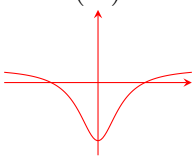
2. Match each graph below with its corresponding function. Note that one function will not have a matching graph.

$$f(x) = \frac{x^2-4}{x^2+1} \qquad g(x) = \frac{x-2}{x+1} \qquad h(x) = \frac{(x+2)(x-1)}{x-2} \qquad k(x) = \frac{1}{x^2-4}$$

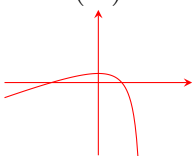
(A)



(B)



(C)

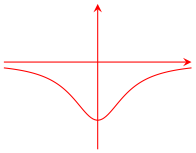


A: $g(x) = \frac{x-2}{x+1}$
B: $f(x) = \frac{x^2-4}{x^2+1}$
C: $h(x) = \frac{(x+2)(x-1)}{x-2}$
Unmatched function: $k(x) = \frac{1}{x^2-4}$

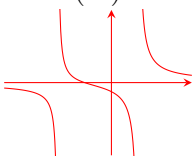
3. Match each graph below with its corresponding function. Note that one function will not have a matching graph.

$$f(x) = \frac{x+1}{(x-1)(x+2)} \qquad g(x) = \frac{(x+1)(x-1)}{x} \qquad h(x) = \frac{-2}{x^2+1} \qquad k(x) = \frac{x^2+1}{x^2-1}$$

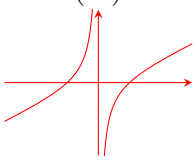
(A)



(B)



(C)

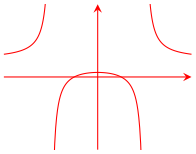


A: $h(x) = \frac{-2}{x^2+1}$
B: $f(x) = \frac{x+1}{(x-1)(x+2)}$
C: $g(x) = \frac{(x+1)(x-1)}{x}$
Unmatched function: $k(x) = \frac{x^2+1}{x^2-1}$

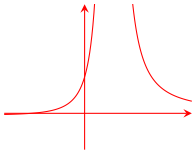
4. Match each graph below with its corresponding function. Note that one function will not have a matching graph.

$$f(x) = \frac{x-3}{x-1} \qquad g(x) = \frac{x^2-1}{x^2-4} \qquad h(x) = \frac{x+2}{(x-1)^2} \qquad k(x) = \frac{2x+1}{x+2}$$

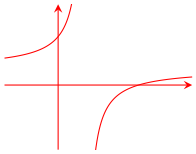
(A)



(B)



(C)



A: $g(x) = \frac{x^2-1}{x^2-4}$
B: $h(x) = \frac{x+2}{(x-1)^2}$
C: $f(x) = \frac{x-3}{x-1}$
Unmatched function: $k(x) = \frac{2x+1}{x+2}$

5. Consider the rational function

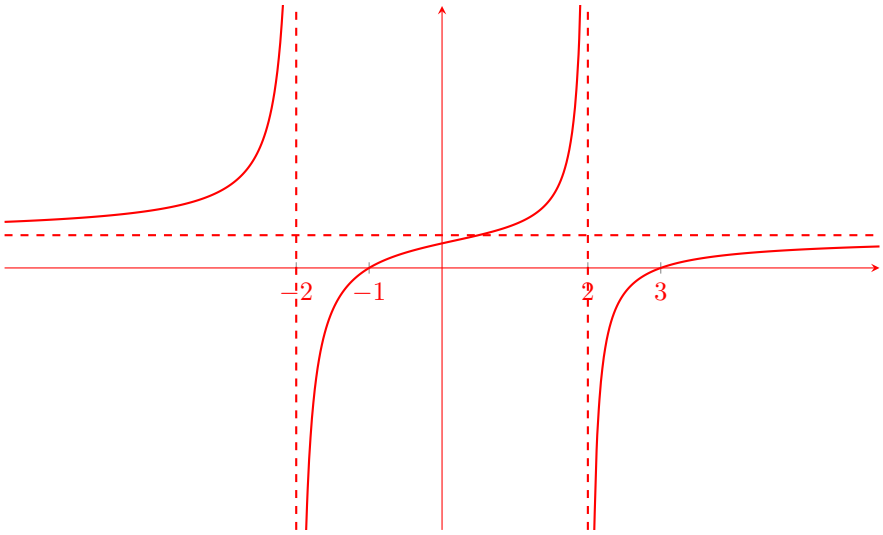
$$r(x) = \frac{(x-3)(x+1)}{(x-2)(x+2)}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

Zeros at $x = -1, 3$. Vertical asymptotes at $x = -2, 2$. y-intercept: $r(0) = \frac{(-3)(1)}{(-2)(2)} = \frac{3}{4}$.

Expanding numerator and denominator: $r(x) = \frac{x^2-2x-3}{x^2-4}$. As $x \to \pm\infty$, $r(x) \approx \frac{x^2}{x^2} = 1$, so the Horizontal Asymptote is $y = 1$.

	$(x-3)$	$(x+1)$	$(x-2)$	$(x+2)$	Total Sign
$x < -2$	$-$	$-$	$-$	$-$	$+$
$-2 < x < -1$	$-$	$-$	$-$	$+$	$-$
$-1 < x < 2$	$-$	$+$	$-$	$+$	$+$
$2 < x < 3$	$-$	$+$	$+$	$+$	$-$
$3 < x$	$+$	$+$	$+$	$+$	$+$



6. Consider the rational function

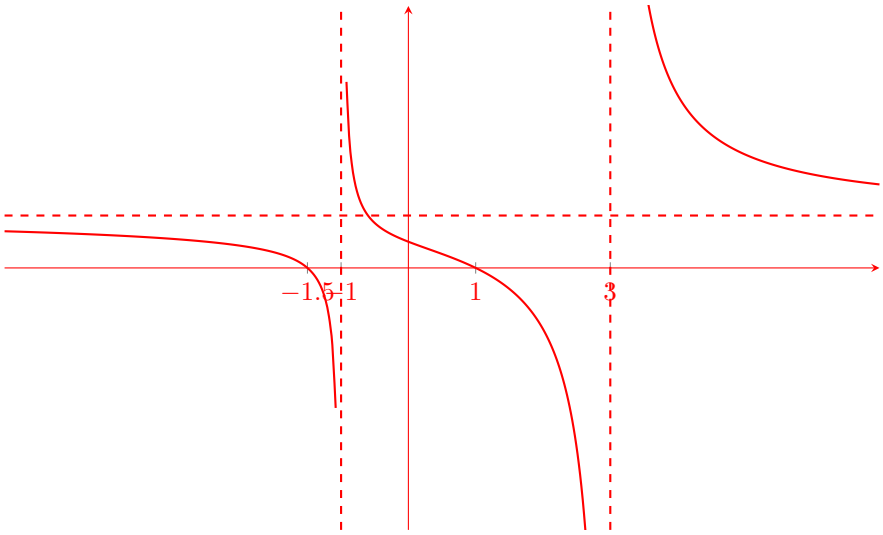
$$r(x) = \frac{(2x+3)(x-1)}{(x+1)(x-3)}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

Zeros at $x = -1.5, 1$. Vertical asymptotes at $x = -1, 3$. y-intercept: $r(0) = \frac{(3)(-1)}{(1)(-3)} = 1$.

Expanding numerator and denominator: $r(x) = \frac{2x^2+x-3}{x^2-2x-3}$. As $x \to \pm\infty$, $r(x) \approx \frac{2x^2}{x^2} = 2$, so the Horizontal Asymptote is $y = 2$.

	$(2x+3)$	$(x-1)$	$(x+1)$	$(x-3)$	Total Sign
$x < -1.5$	$-$	$-$	$-$	$-$	$+$
$-1.5 < x < -1$	$+$	$-$	$-$	$-$	$-$
$-1 < x < 1$	$+$	$-$	$+$	$-$	$+$
$1 < x < 3$	$+$	$+$	$+$	$-$	$-$
$3 < x$	$+$	$+$	$+$	$+$	$+$



7. Consider the rational function

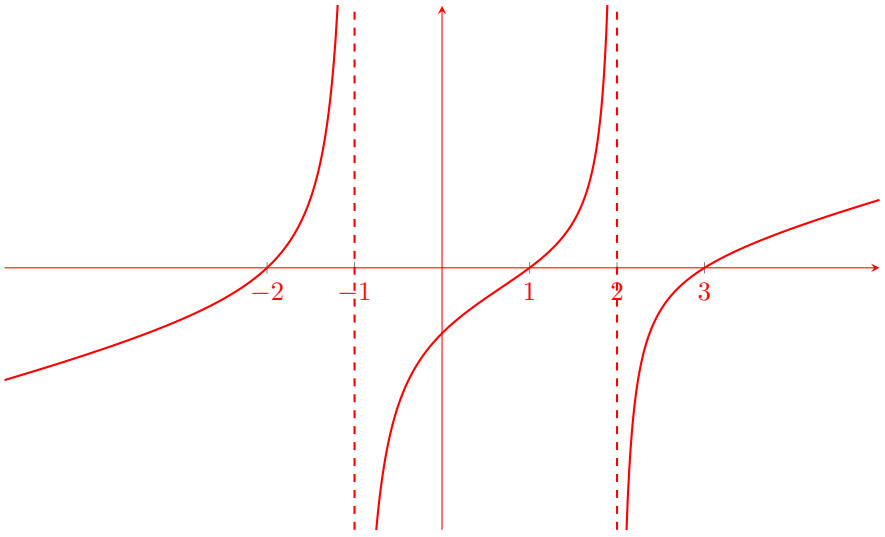
$$r(x) = \frac{(x-1)(x+2)(x-3)}{(x+1)(x-2)}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

Zeros at $x = -2, 1, 3$. Vertical asymptotes at $x = -1, 2$. y-intercept: $r(0) = \frac{(-1)(2)(-3)}{(1)(-2)} = -3$.

Expanding numerator and denominator: $r(x) = \frac{x^3-2x^2-5x+6}{x^2-x-2}$. As $x \to \pm\infty$, $r(x) \approx \frac{x^3}{x^2} = x \to \pm\infty$, so there is no Horizontal Asymptote.

	$(x-1)$	$(x+2)$	$(x-3)$	$(x+1)$	$(x-2)$	Total Sign
$x < -2$	$-$	$-$	$-$	$-$	$-$	$-$
$-2 < x < -1$	$-$	$+$	$-$	$-$	$-$	$+$
$-1 < x < 1$	$-$	$+$	$-$	$+$	$-$	$-$
$1 < x < 2$	$+$	$+$	$-$	$+$	$-$	$+$
$2 < x < 3$	$+$	$+$	$-$	$+$	$+$	$-$
$3 < x$	$+$	$+$	$+$	$+$	$+$	$+$



8. Consider the rational function

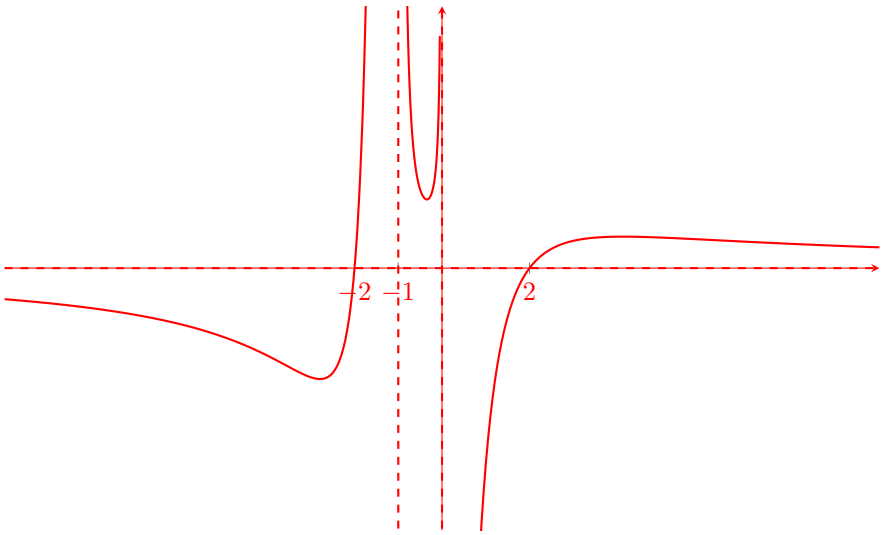
$$r(x) = \frac{(x+2)(x-2)}{x(x+1)^2}.$$

Perform a complete analysis of $r(x)$ by finding its intercepts, asymptotes, and sign behavior, then sketch its graph. Do not use a calculator or any graphing software to complete this problem.

Zeros at $x = -2, 2$. Vertical asymptotes at $x = -1, 0$. No y-intercept because $r(0)$ is undefined.

Expanding numerator and denominator: $r(x) = \frac{x^2-4}{x^3+2x^2+x}$. As $x \to \pm\infty$, $r(x) \approx \frac{x^2}{x^3} = \frac{1}{x} \to 0$, so the Horizontal Asymptote is $y = 0$.

	$(x+2)$	$(x-2)$	x	$(x+1)^2$	Total Sign
$x < -2$	$-$	$-$	$-$	$+$	$-$
$-2 < x < -1$	$+$	$-$	$-$	$+$	$+$
$-1 < x < 0$	$+$	$-$	$-$	$+$	$+$
$0 < x < 2$	$+$	$-$	$+$	$+$	$-$
$2 < x$	$+$	$+$	$+$	$+$	$+$



9. Classify each of the following functions by listing all applicable labels from: **Power function**, **Polynomial**, and **Rational function**.

   (a) $f(x) = 7x^5$

   Power function, Polynomial, Rational function

   (b) $f(x) = 4x^{-3}$

   Power function, Rational function

   (c) $f(x) = 3x^{2.5}$

   Power function

   (d) $f(x) = 2x^4 - 5x^2 + 1$

   Polynomial, Rational function

   (e) $f(x) = \frac{x^2-4}{x+3}$

   Rational function

   (f) $f(x) = \frac{5}{x^3+2x}$

   Rational function

   (g) $f(x) = \frac{1}{2}x^8$

   Power function, Polynomial, Rational function

   (h) $f(x) = -9$

   Power function, Polynomial, Rational function

   (i) $f(x) = 6x^{-1/2}$

   Power function

   (j) $f(x) = \frac{3x^3+1}{4}$

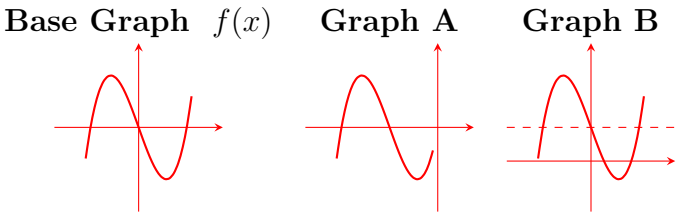
   Polynomial, Rational function

Function Transformations

1. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = f(x+2), \qquad \text{(ii) } y = f(x-2), \qquad \text{(iii) } y = f(x)+2$$

Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)

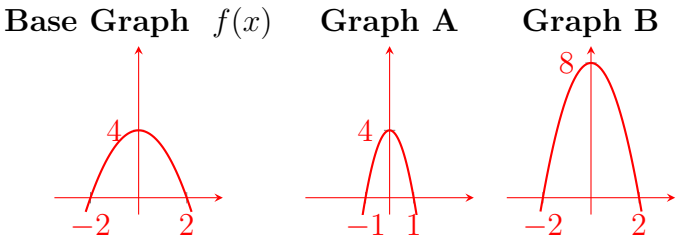


Graph A: (i) $y = f(x+2)$ (shifted left 2 units)
Graph B: (iii) $y = f(x)+2$ (shifted up 2 units)
Unmatched formula: (ii) $y = f(x-2)$

2. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = f(2x), \qquad \text{(ii) } y = 2f(x), \qquad \text{(iii) } y = f\left(\frac{1}{2}x\right)$$

Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)

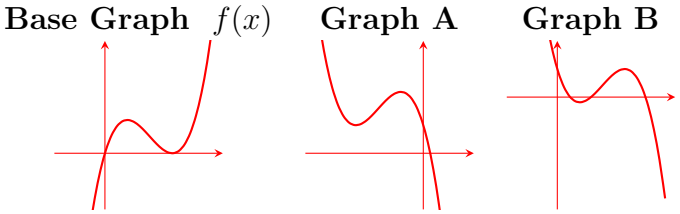


Graph A: (i) $y = f(2x)$ (horizontally compressed by a factor of 2; x-intercepts move to $x = \pm 1$)
Graph B: (ii) $y = 2f(x)$ (vertically stretched by a factor of 2; y-intercept moves to $y = 8$)
Unmatched formula: (iii) $y = f\left(\frac{1}{2}x\right)$

3. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = -f(x) + 1, \qquad \text{(ii) } y = f(-x) + 1, \qquad \text{(iii) } y = -f(x+1)$$

Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)

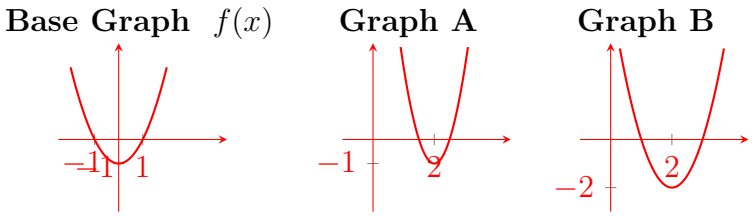


Graph A: (ii) $y = f(-x) + 1$ (reflected across the y-axis, then shifted up 1 unit)
Graph B: (i) $y = -f(x) + 1$ (reflected across the x-axis, then shifted up 1 unit)
Unmatched formula: (iii) $y = -f(x+1)$

4. The graph of a function $f(x)$ is shown below. Consider the following three transformation formulas:

$$\text{(i) } y = f(2x-4), \qquad \text{(ii) } y = f(2x+4), \qquad \text{(iii) } y = 2f(x-2)$$

Match Graph A and Graph B with their corresponding formulas. (One formula will not match either graph.)

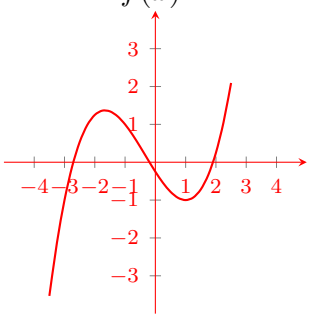


Graph A: (i) $y = f(2x-4) = f(2(x-2))$ (horizontally compressed by 2, then shifted right 2; vertex at $(2, -1)$)
Graph B: (iii) $y = 2f(x-2)$ (shifted right 2, then vertically stretched by 2; vertex at $(2, -2)$)
Unmatched formula: (ii) $y = f(2x+4)$

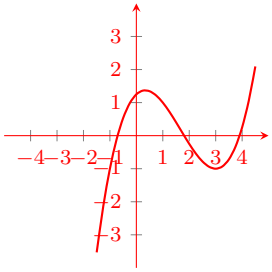
5. For each problem below, graph the given transformation of $f(x)$. If multiple transformations are involved, draw each step on a separate set of axes, as well as your final answer.

(a) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(x-2)+1$.

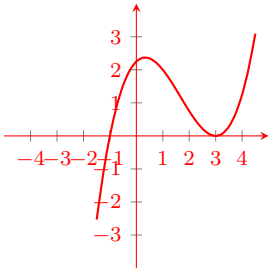
$f(x)$:



Step 1: Shift right 2 units $(y = f(x-2))$

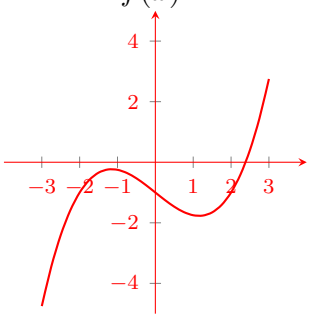


Step 2 / Final Answer: Shift up 1 unit $(g(x) = f(x-2)+1)$

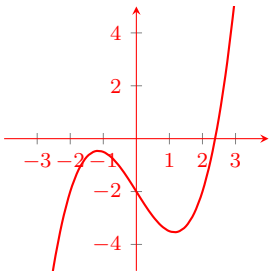


(b) Given the graph of $f(x)$ below, graph the transformed function $g(x) = -2f(x)$.

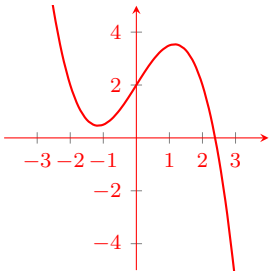
$f(x)$:



Step 1: Vertical stretch by a factor of 2 $(y = 2f(x))$

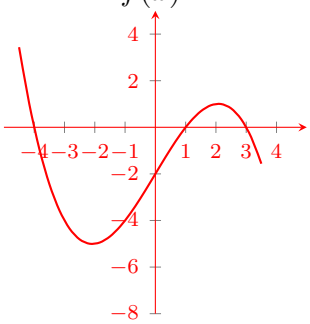


Step 2 / Final Answer: Reflect across the x-axis $(g(x) = -2f(x))$

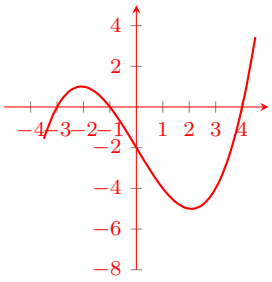


(c) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(-x) - 2$.

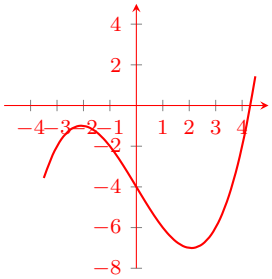
$f(x)$:



Step 1: Reflect across the y-axis $(y = f(-x))$

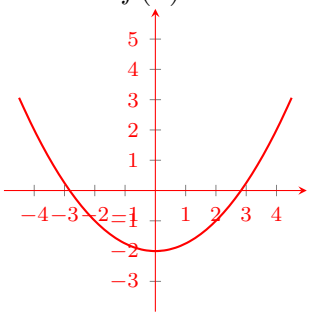


Step 2 / Final Answer: Shift down 2 units $(g(x) = f(-x) - 2)$

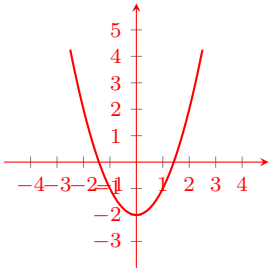


(d) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(2x) + 3$.

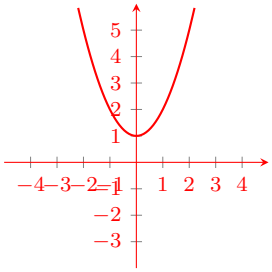
$f(x)$:



Step 1: Horizontal compression by a factor of 2 $(y = f(2x))$

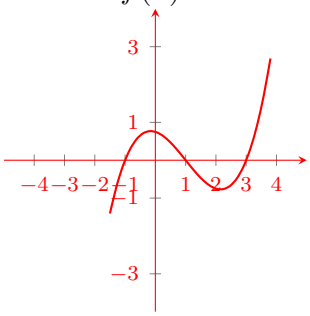


Step 2 / Final Answer: Shift up 3 units $(g(x) = f(2x) + 3)$

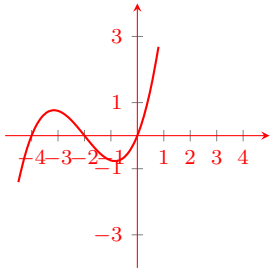


(e) Given the graph of $f(x)$ below, graph the transformed function $g(x) = -f(x+3)$.

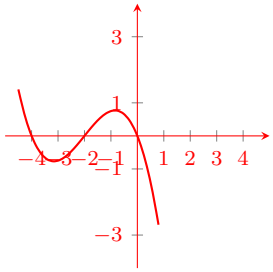
$f(x)$:



Step 1: Shift left 3 units $(y = f(x+3))$

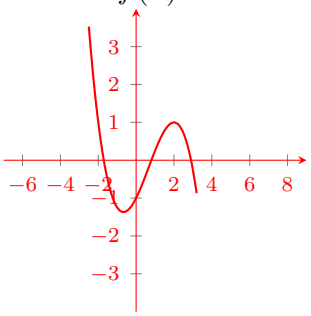


Step 2 / Final Answer: Reflect across the x-axis $(g(x) = -f(x+3))$



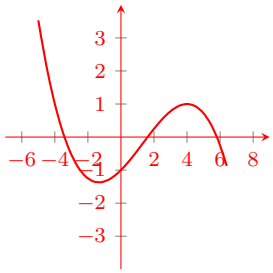
(f) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f\left(\frac{1}{2}x - 1\right)$.

$f(x)$:

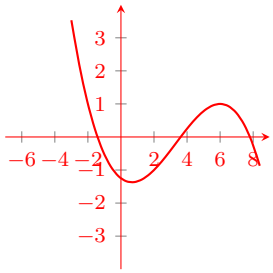


First rewrite $g(x) = f\left(\frac{1}{2}(x-2)\right)$.

Step 1: Horizontal stretch by a factor of 2 $(y = f\left(\frac{1}{2}x\right))$

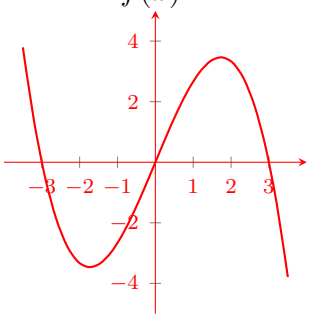


Step 2 / Final Answer: Shift right 2 units $(g(x) = f\left(\frac{1}{2}(x-2)\right))$

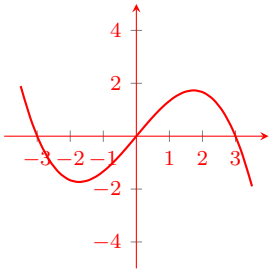


(g) Given the graph of $f(x)$ below, graph the transformed function $g(x) = -\frac{1}{2}f(x)+2$.

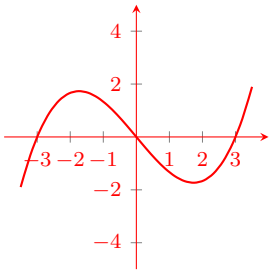
$f(x)$:



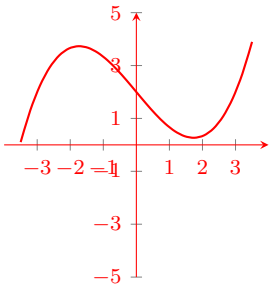
Step 1: Vertical compression by a factor of $\frac{1}{2}$ $(y = \frac{1}{2}f(x))$



Step 2: Reflect across the x-axis $(y = -\frac{1}{2}f(x))$

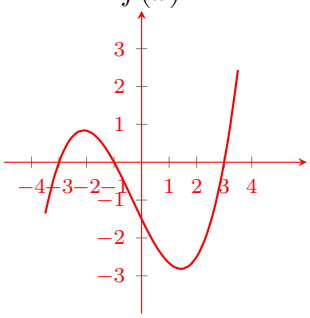


Step 3 / Final Answer: Shift up 2 units $(g(x) = -\frac{1}{2}f(x) + 2)$



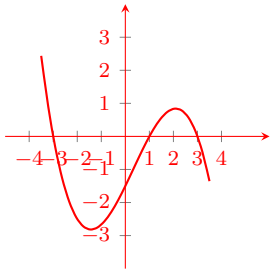
(h) Given the graph of $f(x)$ below, graph the transformed function $g(x) = f(-x+2)$.

$f(x)$:

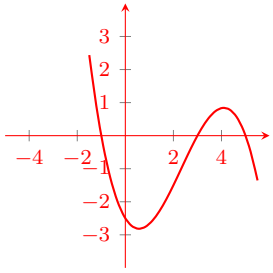


First rewrite $g(x) = f(-(x-2))$.

Step 1: Reflect across the y-axis $(y = f(-x))$



Step 2 / Final Answer: Shift right 2 units $(g(x) = f(-(x-2)))$



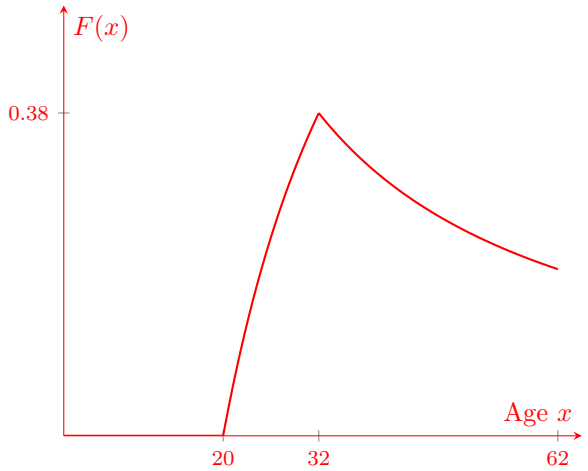
Modeling

1. Suppose a person enters medical school and specialized residency training at age 20, completes all training at age 32, and practices medicine until retiring at age 62. Let x represent the person's age in years ($0 \leq x \leq 62$).

   Write a piecewise formula for $F(x)$, the proportion of their total lifetime lived so far that they spent in medical training. Then sketch a graph of $F(x)$ vs. x.

   Formula for $F(x)$:

$$F(x) = \begin{cases} 0 & \text{if } 0 \leq x < 20, \\ \frac{x-20}{x} & \text{if } 20 \leq x \leq 32, \\ \frac{12}{x} & \text{if } 32 < x \leq 62. \end{cases}$$



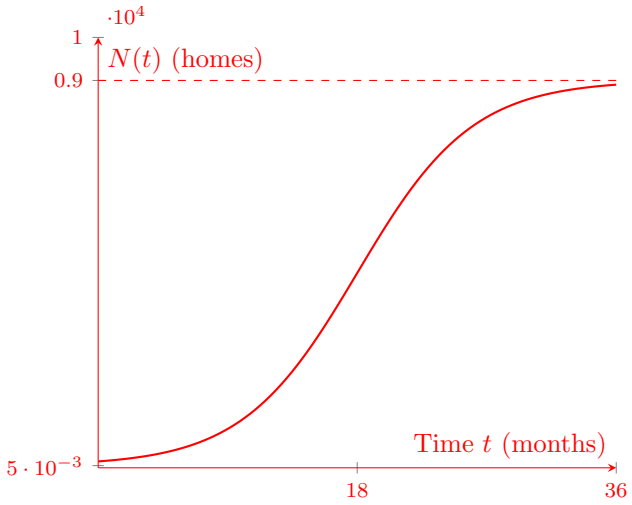
2. A suburban township with 10,000 residential homes launches a community solar adoption initiative. Market research reveals that:

   (i) 10% of the homes have heavily shaded roofs and can never install solar panels,

   (ii) at time $t = 0$ months, exactly 50 early-adopter homes have solar panels installed,

   (iii) the promotion campaign runs for 36 months, after which adoption levels off.

   Draw a possible graph of $N(t)$, the total number of homes with solar panels t months after the launch. Label your axes with appropriate scales and write a short paragraph justifying the shape of your graph.



Initially, adoption grows slowly as awareness spreads from a small base of 50 homes. As neighborhood visibility increases, the rate of new installations accelerates. Eventually, as the pool of unequipped eligible homes shrinks toward the effective cap of 9,000 homes (90% of 10,000), the growth rate decelerates, causing the curve to approach a horizontal asymptote at $N = 9,000$.

3. A community climbing gym offers a pay-per-visit rate of \$15 per visit. Alternatively, climbers can purchase a monthly pass for a flat fee of \$40 plus a reduced rate of \$5 per visit.

   (a) Calculate the total monthly cost under both options if a climber visits 2 times in a month. What if they visit 8 times?

   2 visits: Pay-per-visit $= 2 \times 15 = \$30$; Monthly pass $= 40 + 2(5) = \$50$. (Pay-per-visit is \$20 cheaper).
   8 visits: Pay-per-visit $= 8 \times 15 = \$120$; Monthly pass $= 40 + 8(5) = \$80$. (Monthly pass is \$40 cheaper).

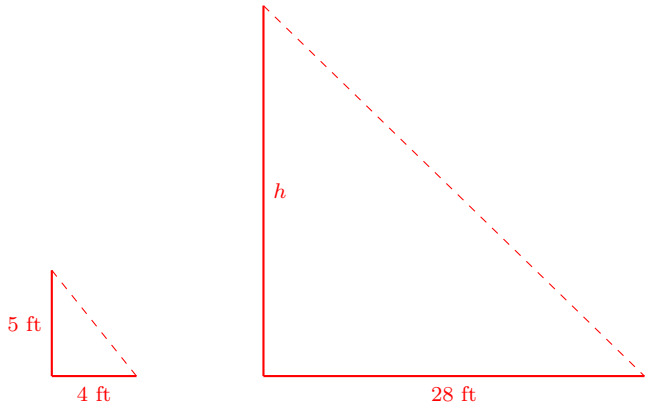
   (b) Write a formula for the monthly savings $S(v)$ (in dollars) gained by purchasing the monthly pass instead of paying per visit, as a function of the number of visits v.

   $S(v) = 15v - (40 + 5v) = 10v - 40$

   (c) Find the minimum number of visits v in a month required for the monthly pass to be more cost-effective than paying per visit.

   Set $S(v) > 0 \implies 10v - 40 > 0 \implies v > 4$.
   A climber must visit at least 5 times in a month for the pass to save money.

4. To estimate the height of a tall oak tree in a park, a student measures the length of the tree's shadow on level ground to be 28 feet. At the exact same time, a 5-foot tall wooden fence post nearby casts a shadow of length 4 feet. Draw a sketch representing this situation, and find the height of the oak tree.



Setting up a proportion using similar triangles:

$$\frac{h}{28} = \frac{5}{4} \implies h = 28 \times \frac{5}{4} = 35 \text{ feet.}$$

5. The illuminance I (measured in lux) produced by a light bulb on a surface is inversely proportional to the square of the distance d (in meters) from the light source.

   (a) Write a general formula expressing I in terms of d.

   $I = \frac{k}{d^2}$

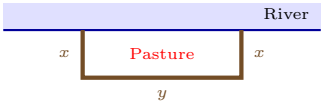
   (b) If the illuminance is 180 lux at a distance of 2 meters, find the constant of proportionality k and state its units.

   $180 = \frac{k}{2^2} \implies k = 180 \times 4 = 720 \text{ lux} \cdot \text{m}^2$.

   (c) Determine the illuminance when the distance from the light bulb is increased to 6 meters.

   $I = \frac{720}{6^2} = \frac{720}{36} = 20 \text{ lux}$.

6. A farmer wants to construct a rectangular pasture along a straight riverbank (as shown below). No fencing is required along the river. The farmer has 160 meters of fencing material available for the remaining three sides.



(a) Let x be the length (in meters) of the two sides perpendicular to the river. Express the length y of the side parallel to the river in terms of x.

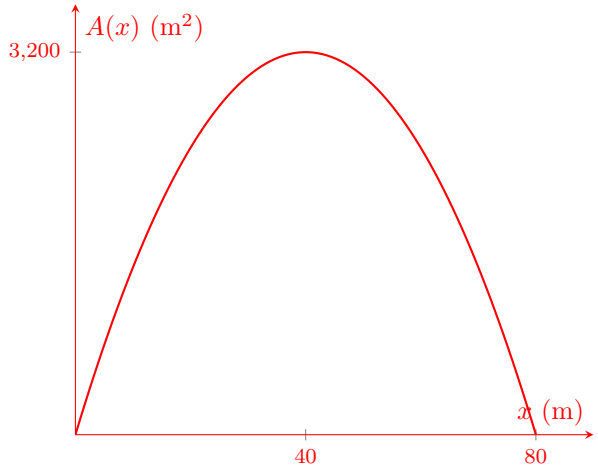
$2x + y = 160 \implies y = 160 - 2x$.

(b) Write a formula for $A(x)$, the enclosed area of the pasture, as a function of x.

$A(x) = x \cdot y = x(160 - 2x) = -2x^2 + 160x$.

(c) Find the dimensions x and y that maximize the area of the pasture, and calculate the maximum area. (No calculus here: just use symmetry.)

The area function $A(x) = -2x(x - 80)$ has zeros at $x = 0$ and $x = 80$.
By symmetry, the maximum area occurs at $x = 40$ meters.
The corresponding length parallel to the river is $y = 160 - 2(40) = 80$ meters.
Maximum area $= A(40) = 40 \times 80 = 3,200$ m^2.



7. A comedy club with a capacity of 300 seats charges \$25 per ticket and consistently sells out every show. Market research indicates that for every \$1 increase in ticket price above \$25, attendance decreases by 10 people.

   (a) Find a linear formula for $N(p)$, the number of attendees, as a function of the ticket price p (where $p \geq 25$).

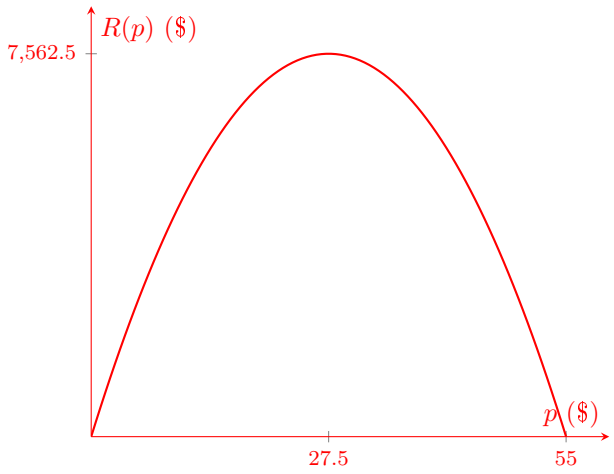
   $N(p) - 300 = -10(p - 25) \implies N(p) = -10p + 550.$

   (b) Write a formula for total revenue $R(p)$ per show as a function of ticket price p.

   $R(p) = p \cdot N(p) = p(-10p + 550) = -10p^2 + 550p.$

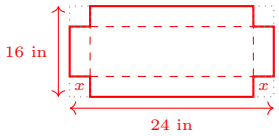
   (c) Determine the ticket price p that maximizes revenue per show, find the maximum total revenue, and sketch a graph of $R(p)$ vs. p. (No calculus here: just use symmetry.)

   The zeros of $R(p)$ are $p = 0$ and $p = 55$.
   By symmetry, maximum revenue occurs at $p = \frac{0+55}{2} = \$27.50$.
   Maximum revenue $= R(27.50) = -10(27.50)^2 + 550(27.50) = \$7,562.50$.



8. **Challenge Problem**

An open-top box is to be constructed from a rectangular piece of cardboard measuring 16 inches by 24 inches by cutting equal squares of side length x inches from each of the four corners and folding up the resulting flaps (as shown below).



(a) Write expressions for the height, width, and length of the box in terms of x.

Height $= x$ in, Width $= 16 - 2x$ in, Length $= 24 - 2x$ in.

(b) Write a formula for $V(x)$, the volume of the box (in cubic inches), as a function of x.

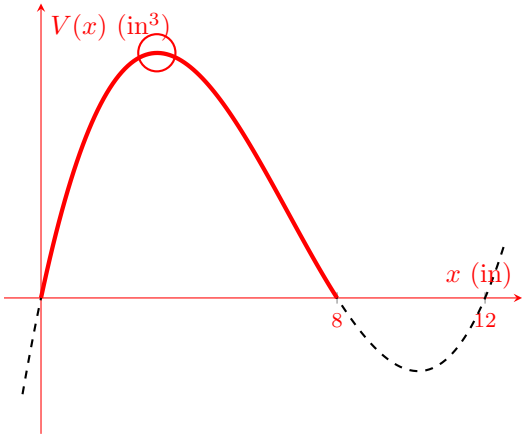
$V(x) = x(16 - 2x)(24 - 2x) = 4x(8 - x)(12 - x) = 4x^3 - 80x^2 + 384x$.

(c) State the practical physical domain for x in this context.

All dimensions must be strictly positive: $x > 0$, $16 - 2x > 0 \implies x < 8$, and $24 - 2x > 0 \implies x < 12$.

Taking the most restrictive bound gives the practical domain: $0 < x < 8$ inches.

(d) Draw the entire graph of $V(x)$, highlight the portion corresponding to the practical domain, and circle the point where $V(x)$ is maximized.



Rates of Change and Derivative at a Point

For all problems in this section, you may not use any calculus tools beyond what was covered up until Worksheet 4-3. For example, using the power rule, product rule, quotient rule, or chain rule is not allowed.

1. The position $s(t)$ (in feet) of a vehicle traveling along a straight track is recorded at various times t (in seconds) in the table below:

t (sec)	0	1.0	2.0	3.0	4.0
$s(t)$ (ft)	10.0	24.5	51.0	89.5	140.0

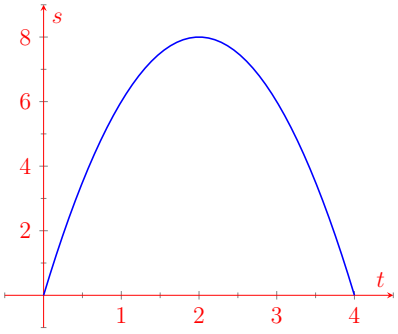
   (a) Calculate the average velocity of the vehicle between $t = 1.0$ and $t = 3.0$ seconds.

$$\text{Average Velocity} = \frac{s(3) - s(1)}{3 - 1} = \frac{89.5 - 24.5}{2} = \frac{65.0}{2} = 32.5 \text{ ft/sec}$$

   (b) Calculate the average velocity of the vehicle between $t = 2.0$ and $t = 4.0$ seconds.

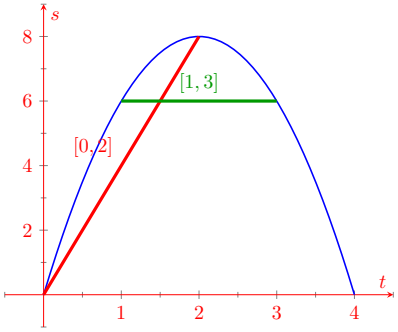
$$\text{Average Velocity} = \frac{s(4) - s(2)}{4 - 2} = \frac{140.0 - 51.0}{2} = \frac{89.0}{2} = 44.5 \text{ ft/sec}$$

2. Given the graph of the position function $s(t) = -2(t-2)^2 + 8$ below, draw and label lines whose slopes correspond to the average velocity over the specified time intervals:

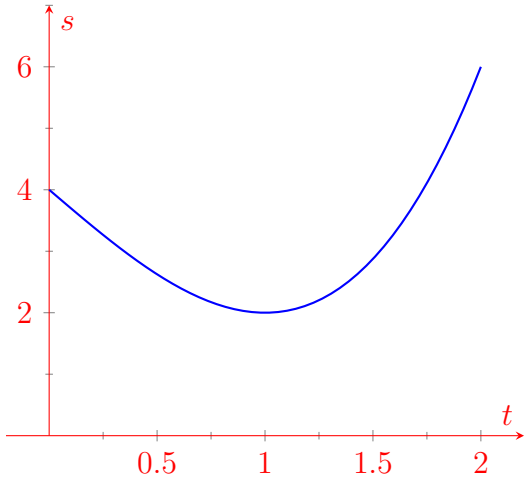


   (a) Average velocity on the interval $[0, 2]$.

   (b) Average velocity on the interval $[1, 3]$.

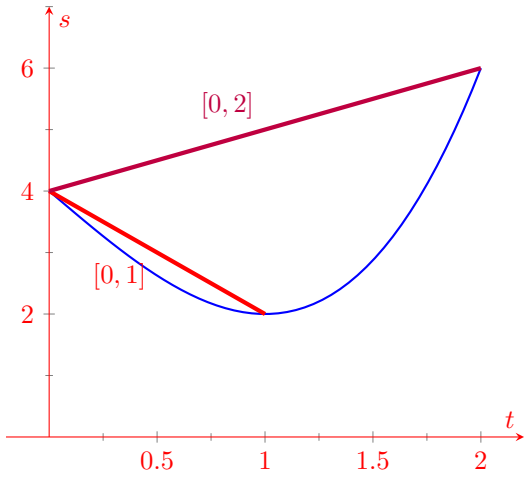


3. Given the graph of the position function $s(t) = t^3 - 3t + 4$ below, draw and label lines whose slopes correspond to the average velocity over the specified time intervals:



(a) Average velocity on the interval $[0, 1]$.

(b) Average velocity on the interval $[0, 2]$.



4. The table below provides values of the average velocity $\frac{s(3+h)-s(3)}{h}$ of an object near $t = 3$ seconds:

h (sec)	Avg. Vel on $[3, 3+h]$ (m/s)	h (sec)	Avg. Vel on $[3, 3+h]$ (m/s)
-0.1	5.90	0.1	6.10
-0.01	5.99	0.01	6.01
-0.001	5.999	0.001	6.001

Estimate the instantaneous velocity of the object at $t = 3$ seconds.

Estimated Instantaneous Velocity = 6 m/s.

5. The table below provides values of the average velocity $\frac{s(1+h)-s(1)}{h}$ of an object near $t = 1$ second:

h (sec)	Avg. Vel on $[1, 1+h]$ (ft/s)	h (sec)	Avg. Vel on $[1, 1+h]$ (ft/s)
-0.1	-2.85	0.1	-3.15
-0.01	-2.985	0.01	-3.015
-0.001	-2.9985	0.001	-3.0015

Estimate the instantaneous velocity of the object at $t = 1$ second.
Estimated Instantaneous Velocity $= -3$ ft/s.

6. For each of the following position functions $s(t)$ (where s is in meters and t is in seconds), compute the exact instantaneous velocity at the given time using the limit definition $v(a) = \lim_{h\to 0} \frac{s(a+h)-s(a)}{h}$.

(a) $s(t) = 2t^2 - 3t$ at $t = 2$.

$$\begin{aligned}\lim_{h\to 0} \frac{s(2+h)-s(2)}{h} &= \lim_{h\to 0} \frac{[2(2+h)^2 - 3(2+h)] - [2(2)^2 - 3(2)]}{h} \\ &= \lim_{h\to 0} \frac{2(4+4h+h^2) - 6 - 3h - 2}{h} = \lim_{h\to 0} \frac{8+8h+2h^2-8-3h}{h} \\ &= \lim_{h\to 0} \frac{5h+2h^2}{h} = \lim_{h\to 0} (5+2h) = 5 \text{ m/s}\end{aligned}$$

(b) $s(t) = -t^2 + 5t + 1$ at $t = 3$.

$$\begin{aligned}\lim_{h\to 0} \frac{s(3+h)-s(3)}{h} &= \lim_{h\to 0} \frac{[-(3+h)^2 + 5(3+h) + 1] - [-(3)^2 + 5(3) + 1]}{h} \\ &= \lim_{h\to 0} \frac{-(9+6h+h^2) + 15 + 5h + 1 - 7}{h} = \lim_{h\to 0} \frac{-h-h^2}{h} \\ &= \lim_{h\to 0} (-1-h) = -1 \text{ m/s}\end{aligned}$$

(c) $s(t) = 3t^2 + 4$ at $t = 1$.

$$\begin{aligned}\lim_{h\to 0} \frac{s(1+h)-s(1)}{h} &= \lim_{h\to 0} \frac{[3(1+h)^2 + 4] - [3(1)^2 + 4]}{h} \\ &= \lim_{h\to 0} \frac{3(1+2h+h^2) + 4 - 7}{h} = \lim_{h\to 0} \frac{6h+3h^2}{h} \\ &= \lim_{h\to 0} (6+3h) = 6 \text{ m/s}\end{aligned}$$

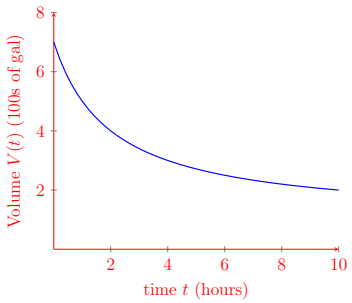
(d) $s(t) = t^3$ at $t = 2$.

$$\begin{aligned}\lim_{h\to 0} \frac{s(2+h)-s(2)}{h} &= \lim_{h\to 0} \frac{(2+h)^3 - 2^3}{h} = \lim_{h\to 0} \frac{(8+12h+6h^2+h^3) - 8}{h} \\ &= \lim_{h\to 0} \frac{12h+6h^2+h^3}{h} = \lim_{h\to 0} (12+6h+h^2) = 12 \text{ m/s}\end{aligned}$$

(e) $s(t) = \frac{2}{t}$ at $t = 2$.

$$\lim_{h\to 0} \frac{s(2+h) - s(2)}{h} = \lim_{h\to 0} \frac{\frac{2}{2+h} - 1}{h} = \lim_{h\to 0} \frac{\frac{2-(2+h)}{2+h}}{h}$$
$$= \lim_{h\to 0} \frac{-h}{h(2+h)} = \lim_{h\to 0} \frac{-1}{2+h} = -\frac{1}{2} \text{ m/s}$$

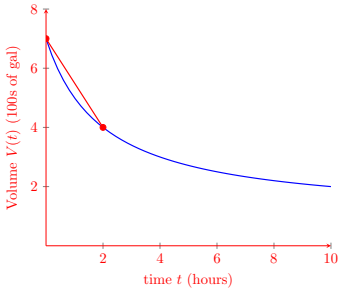
7. A water storage tank is being drained for maintenance. The volume of water remaining in the tank (in hundreds of gallons) t hours after draining begins is modeled by $V(t) = \frac{12}{t+2} + 1$.



(a) How many hundreds of gallons of water were drained during the first 2 hours? What was the average rate of draining per hour over this period?

$V(0) = 7$, $V(2) = 4$. Water drained $= 7 - 4 = 3$ (300 gallons). Average rate $= \frac{V(2)-V(0)}{2-0} = \frac{4-7}{2} = -1.5$ hundred gal/hr.

(b) Illustrate the average rate of change on the graph above.



(c) Rank the following quantities in order from smallest to largest:

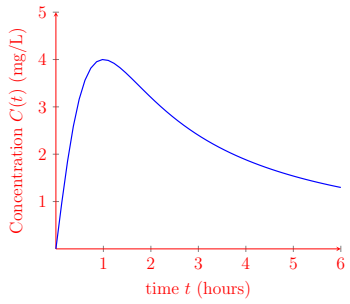
(i) Instantaneous rate of change of volume at $t = 0$.

(ii) Average rate of change of volume between $t = 0$ and $t = 2$.

(iii) Instantaneous rate of change of volume at $t = 2$.

(i) < (ii) < (iii) (since all are negative, the steepest negative slope is the smallest/most negative: $-3 < -1.5 < -0.75$).

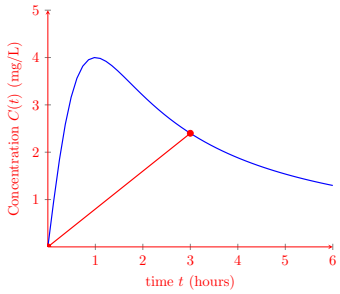
8. The concentration of a medication in a patient's bloodstream (in mg/L) t hours after ingestion is modeled by $C(t) = \frac{8t}{t^2+1}$.



(a) Find the average rate of change of medication concentration over the first 3 hours ($[0, 3]$).

$C(0) = 0$, $C(3) = 2.4$. Average RoC $= \frac{2.4-0}{3-0} = 0.8$ mg/L per hour.

(b) Illustrate the average rate of change on the graph above.



(c) Rank the following quantities in order from smallest to largest:

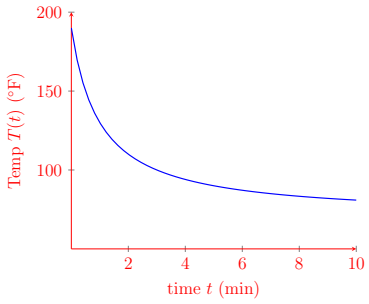
(i) Instantaneous rate of change at $t = 1$.

(ii) Average rate of change on $[0, 3]$.

(iii) Instantaneous rate of change at $t = 3$.

(iii) < (i) < (ii) (at $t = 1$, tangent slope is 0; at $t = 3$, tangent slope is negative; average slope on $[0, 3]$ is positive).

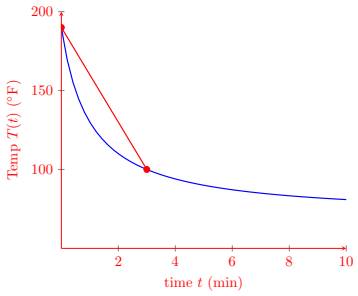
9. The temperature (in °F) of a freshly baked pie placed on a cooling rack t minutes after coming out of the oven is given by $T(t) = 70 + \frac{120}{t+1}$.



(a) Find the average cooling rate (in °F/min) of the pie over the first 3 minutes.

$T(0) = 190$, $T(3) = 100$. Average rate $= \frac{100-190}{3-0} = -30$°F/min.

(b) Illustrate the average cooling rate on the graph above.



(c) Estimate $T'(3)$ using $h = 0.001$.

$\frac{T(3.001)-T(3)}{0.001} = \frac{(70+29.9925)-100}{0.001} \approx -7.5$°F/min.

10. Consider the function $f(x) = x^2 - 4x$.

(a) Fill in the table below to estimate $f'(3)$:

h	Avg. RoC on $[3, 3+h]$	h	Avg. RoC on $[3, 3+h]$
-0.1	1.9	0.1	2.1
-0.01	1.99	0.01	2.01
-0.001	1.999	0.001	2.001

(b) Find $f'(3)$ exactly using the limit definition of the derivative.

$$\begin{aligned} f'(3) &= \lim_{h\to 0} \frac{f(3+h) - f(3)}{h} \\ &= \lim_{h\to 0} \frac{[(3+h)^2 - 4(3+h)] - [3^2 - 4(3)]}{h} \\ &= \lim_{h\to 0} \frac{(9 + 6h + h^2 - 12 - 4h) - (-3)}{h} \\ &= \lim_{h\to 0} \frac{9 + 2h + h^2 - 12 + 3}{h} \\ &= \lim_{h\to 0} \frac{2h + h^2}{h} \\ &= \lim_{h\to 0} \frac{h(2+h)}{h} \\ &= \lim_{h\to 0} (2+h) = 2 \end{aligned}$$

(c) Find the equation of the line tangent to the graph of $y = f(x)$ at $x = 3$.

Point $(3, -3)$, slope $m = 2 \implies y - (-3) = 2(x - 3) \implies y = 2x - 9$.

11. Consider the function $f(x) = -2x^2 + 5$.

(a) Fill in the table below to estimate $f'(1)$:

h	Avg. RoC on $[1, 1+h]$	h	Avg. RoC on $[1, 1+h]$
-0.1	-3.8	0.1	-4.2
-0.01	-3.98	0.01	-4.02
-0.001	-3.998	0.001	-4.002

(b) Find $f'(1)$ exactly using the limit definition of the derivative.

$$\begin{aligned} f'(1) &= \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h\to 0} \frac{[-2(1+h)^2 + 5] - [-2(1)^2 + 5]}{h} \\ &= \lim_{h\to 0} \frac{[-2(1+2h+h^2) + 5] - 3}{h} \\ &= \lim_{h\to 0} \frac{-2 - 4h - 2h^2 + 5 - 3}{h} \\ &= \lim_{h\to 0} \frac{-4h - 2h^2}{h} \\ &= \lim_{h\to 0} \frac{h(-4 - 2h)}{h} \\ &= \lim_{h\to 0} (-4 - 2h) = -4 \end{aligned}$$

(c) Find the equation of the line tangent to the graph of $y = f(x)$ at $x = 1$.

Point $(1, 3)$, slope $m = -4 \implies y - 3 = -4(x - 1) \implies y = -4x + 7$.

12. Consider the function $f(x) = x^3 - 2x$.

(a) Fill in the table below to estimate $f'(2)$:

h	Avg. RoC on $[2, 2+h]$	h	Avg. RoC on $[2, 2+h]$
-0.1	9.41	0.1	10.61
-0.01	9.9401	0.01	10.0601
-0.001	9.994	0.001	10.006

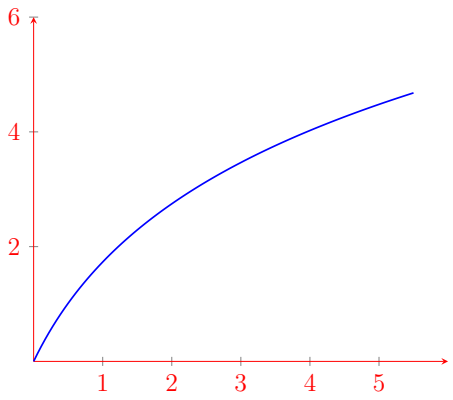
(b) Find $f'(2)$ exactly using the limit definition of the derivative.

$$\begin{aligned} f'(2) &= \lim_{h\to 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h\to 0} \frac{[(2+h)^3 - 2(2+h)] - [2^3 - 2(2)]}{h} \\ &= \lim_{h\to 0} \frac{(8 + 12h + 6h^2 + h^3 - 4 - 2h) - 4}{h} \\ &= \lim_{h\to 0} \frac{10h + 6h^2 + h^3}{h} \\ &= \lim_{h\to 0} \frac{h(10 + 6h + h^2)}{h} \\ &= \lim_{h\to 0} (10 + 6h + h^2) = 10 \end{aligned}$$

(c) Find the equation of the line tangent to the graph of $y = f(x)$ at $x = 2$.

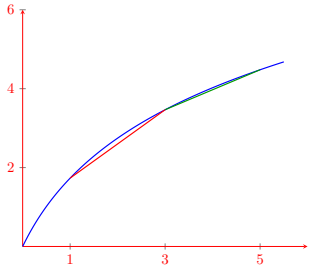
Point $(2, 4)$, slope $m = 10 \implies y - 4 = 10(x - 2) \implies y = 10x - 16$.

13. For the function $f(x)$ graphed below, compare the following pairs of quantities. Indicate which is larger ($>$ or $<$), and illustrate each pair on a copy of the graph:



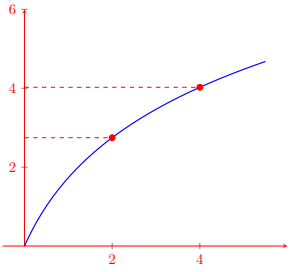
(a) The average rate of change on $[1, 3]$ versus the average rate of change on $[3, 5]$.

Average RoC on $[1, 3]$ > Average RoC on $[3, 5]$ (secant line on $[1, 3]$ is steeper).



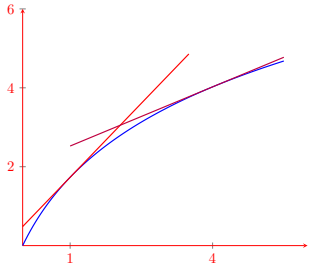
(b) $f(2)$ versus $f(4)$.

$f(2) < f(4)$ (y-value at 4 is higher).

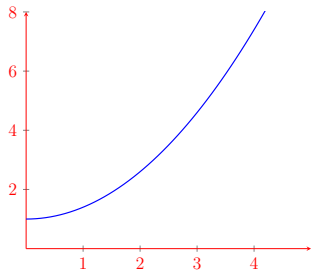


(c) $f'(1)$ versus $f'(4)$.

$f'(1) > f'(4)$ (tangent line at $x = 1$ is steeper).

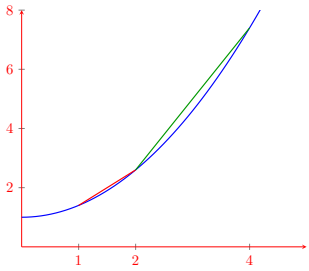


14. For the function $g(x)$ graphed below, compare the following pairs of quantities. Indicate which is larger ($>$ or $<$), and illustrate each pair on a copy of the graph:



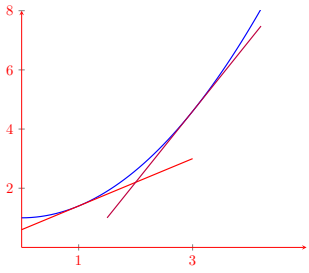
(a) The average rate of change on $[1, 2]$ versus the average rate of change on $[2, 4]$.

Average RoC on $[1, 2]$ $<$ Average RoC on $[2, 4]$ (secant line on $[2, 4]$ is steeper).



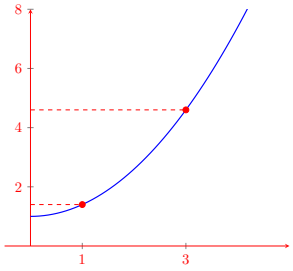
(b) $g'(1)$ versus $g'(3)$.

$g'(1) < g'(3)$ (tangent line at $x = 3$ is steeper).

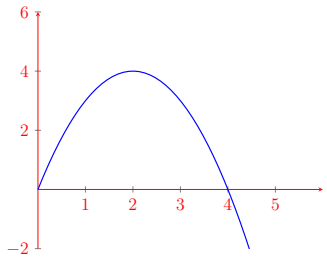


(c) $g(1)$ versus $g(3)$.

$g(1) < g(3)$ (y-value at 3 is higher).

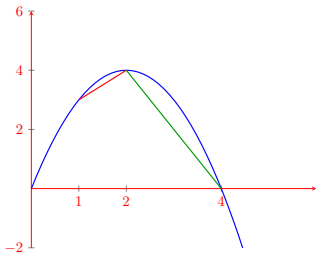


15. For the function $h(x)$ graphed below, compare the following pairs of quantities. Indicate which is larger ($>$ or $<$), and illustrate each pair on a copy of the graph:



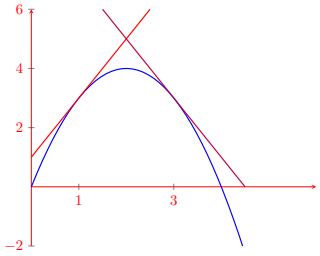
(a) The average rate of change on $[1, 2]$ versus the average rate of change on $[2, 4]$.

Average RoC on $[1, 2]$ > Average RoC on $[2, 4]$.



(b) $h'(1)$ versus $h'(3)$.

$h'(1) > h'(3)$.



(c) $h'(2)$ versus $h'(4)$.

$h'(2) > h'(4)$.

